

Assignment Template

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Preliminaries

Package Installation

Many times you need to install packages, you do that once, but then comment it out:

```
# install.packages("dplyr")
```

Package Loading

```
library(MASS)
```

Assignment

1) *Answer to Problem 1*

This problem involves me saving the number five in a variable called “five”, putting that variable in a data.frame, and then writing that data.frame called “numbersfile.csv” to my working directory.

```
five <- 5
numbers <- data.frame(five)
write.csv(numbers, "numbersfile.csv")
```

2) *Answer to Problem 2*

This problem involves me reading the “numbers.csv” file into R, extracting the five column, and printing it on the screen.

```
numbers <- read.csv("numbersfile.csv")
numbers$five
```

```
## [1] 5
```

3) *Answer to Problem 3 (Example 4.30 from text)*

The three bank tellers can serve customers according to an exponential distribution with parameters, $\lambda = 6$, $\lambda = 6$, and $\lambda = 4$, which refers to the number of customers that they can serve per hour. If Jane comes in, picks one with equal probability, then the total time it would take is distributed according to the PDF:

$$f_X(x) = \frac{2}{3} \cdot 6e^{-6x} + \frac{1}{3} \cdot 4e^{-4x}, \quad x \geq 0$$

Where X is the random variable that measures how long Jane would take.

In order to calculate the expectation of this distribution, we can use the **Moment Generating Function**, which is a transformation of the original PDF. The transform is defined as follows:

$$M(s) = \int_0^{\infty} e^{sx} \left(\frac{2}{3} \cdot 6e^{-6x} + \frac{1}{3} \cdot 4e^{-4x} \right) dx \quad (1)$$

$$= \frac{2}{3} \int_0^{\infty} e^{sx} 6e^{-6x} dx + \frac{1}{3} \int_0^{\infty} e^{sx} 4e^{-4x} dx \quad (2)$$

Now, evaluating the integrals to show my work,

$$\begin{aligned} \int_0^{\infty} e^{sx} 6e^{-6x} dx &= 6 \int_0^{\infty} e^{(s-6)x} dx \\ &= 6 \frac{1}{s-6} e^{(s-6)x} \Big|_0^{\infty} \\ &= 0 - 6 \frac{1}{s-6} e^0 \\ &= \frac{6}{6-s} \end{aligned}$$

Plugging this result back in to equation (2)

$$M(s) = \frac{2}{3} \cdot \frac{6}{6-s} + \frac{1}{3} \cdot \frac{4}{4-s}$$

As a result, the mean of X is just

$$\left. \frac{d}{ds} M(s) \right|_{s=0} = \frac{2}{3} * \frac{6}{(6)^2} + \frac{1}{3} * \frac{4}{(4)^2} = \frac{7}{36}$$

Confirming computationally:

```
mean(2 / 3 * rexp(10000, 6) + 1 / 3 * rexp(10000, 4))
```

```
## [1] 0.1950658
```

So, we will expect Jane to be served in about 11.7 minutes.

4) *Answer to random R question*

```
# First I create a sequence of numbers between -1 and 1.  
x <- seq(-.99,.99, by=.001)  
# I then feed those numbers into f(x)  
  
y<- 2 / (1-x^2)  
  
# Each sequence is then plotted in the plot command,  
# I limit the y axis to be between 0 and 10 for legibility.  
plot(x,y, type="l", ylim=c(0,10))
```

